

# Energetic Electron Injections Inside of GEO Observed by an Operational Constellation

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## Key Points:

- We present the first identifications of substorm injections at low L using LANL-GPS particle data.
- LANL-GPS particle data enables continuous monitoring of inner magnetosphere for injection studies.
- Where coverage alone does not constrain injection region this can be augmented with particle tracing.

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## Abstract

Substorm injections to the inner magnetosphere occur along a broad azimuthal range on the nightside of Earth. This adds difficulty to studying them systematically as most inner magnetosphere observatories have only single-point measurements at any given time or location. The Global Positioning System (GPS) constellation is distributed throughout Medium-Earth Orbit providing an ideal location to probe the inner magnetosphere. The number of satellites and orbital paths available also offers unprecedented coverage in time and space of this region. Our analysis shows that substorm injections can be directly measured by GPS and we present methods for characterizing electron injections. To demonstrate this, we present analysis of electron data from two substorm injection events observed by the energetic particle detectors on the GPS constellation of satellites on 2016-09-27 and 2022-04-10. We are able to distinguish between approximately dispersionless and dispersed injection signatures and estimate the approximate extents of the injection regions.

## 1 Introduction

Substorms are a class of space weather event wherein energy from the solar wind is converted into magnetic flux in the magnetosphere before being converted back into kinetic energy in the form of accelerated and heated plasma (Dungey, 1963). This plasma then interacts with the inner magnetosphere, ionosphere, and upper atmosphere as it moves towards the Earth generating aurora, ionospheric and ground induced currents, and other geospace phenomena. Energetic particle injections from the magnetotail into the inner magnetosphere have long been studied as an important mechanism in substorm development. Much work has been done on the sources and phenomenology of injections as they travel Earthward through the tail (e.g., Gabrielse et al., 2014; Ukhorskiy et al., 2018). After arriving at the inner magnetosphere, numerous studies have investigated the role played by injections on local wave growth (e.g., H. Kim et al., 2021), particle precipitation to the ionosphere (e.g., Sanchez et al., 2022), and charging hazard for satellites in the inner magnetosphere (e.g., Choi et al., 2011; Thomsen et al., 2013; Maldonado et al., 2024).

Several of these studies, perhaps the majority, have been performed using platforms located in geosynchronous orbit (GEO) (e.g., Mauk & Meng, 1983; Reeves et al., 1991; Birn et al., 1998; Rogers et al., 2024). Given the number of platforms located at GEO such as LANL-GEO (Henderson, 2024) and GOES (Dichter et al., 2015; Kress et al., 2020), these studies have contributed significantly to our understanding of injection phenomenology at L-shells of  $\sim 6.6R_E$ . They have not, however been able to elucidate the effects of injections closer to Earth.

Additional insight on this topic has been gained by other observatories following elliptical orbits such as CRRES (e.g., Friedel et al., 1996) and the Van Allen Probes (e.g., H.-J. Kim et al., 2021). Studies using this data have done much to describe the radial structure of the radiation belts (e.g., Reeves et al., 2016), wave-particle interactions (e.g., Noh et al., 2018), etc. However, observations from these platforms have been handicapped by their essentially single-point measurements in azimuthal (or local time) location about the Earth. Due to the nature of the radiation belt environment these missions tend to have relatively short lifespans, i.e., less than ten years in the case of the Van Allen Probes, further restricting their data coverage, albeit in time. Also, given the nature of their orbits, such observations have only a short period of time during which they can sample a given region. A natural consequence is that the ability of such platforms to observe any given example of an energetic particle injection is determined primarily by chance. Furthermore, any conjunctions with other observatories during an injection which might provide more spatial context must, likewise, be owing to serendipity (Turner et al., 2017).

The Global Positioning System (GPS) constellation, which is distributed throughout Medium-Earth Orbit (MEO), is equipped with energetic charged particle sensors supplied by Los Alamos National Laboratory (LANL) (Morley et al., 2017), and is thus ideally placed and reasonably equipped for monitoring of the inner magnetosphere (Morley et al., 2016; Olifer et al., 2024). The 12-hour orbits of the GPS satellites ensure that each platform samples all  $L \gtrsim 4.2$  four times over the course of each orbit. Very few investigations of substorms using LANL-GPS data have been performed: Li et al. (1998) used count rates from three early LANL-GPS satellites to provide context for radiation belt enhancement, measured from low-Earth orbit, as a consequence of intense substorm activity; Ingraham et al. (2001) used early LANL-GPS data to provide context for substorm injection of relativistic electrons, but did not directly examine the injections from the LANL-GPS data; Varotsou et al. (2007) performed a preliminary analysis of electron flux variations at GPS corresponding with sawtooth intervals. To our knowledge, no peer-reviewed literature uses GPS to identify or describe substorm injections. The GPS constellation, then, represents an under-utilized resource for studying energetic electron injections to the inner magnetosphere. In this work we will demonstrate that substorm injections can be directly measured from the GPS constellation using the LANL-GPS particle data, and we present methods for characterizing electron injections using the publicly-available LANL-GPS data.

## 2 LANL-GPS Energetic Charged Particle Data

LANL operates energetic charged particle sensors for environmental monitoring on the GPS constellation. There are currently more than 30 operational GPS satellites, and the most recent LANL-GPS data release (v1.10; Morley et al., 2023) provides data from 27 satellites equipped with LANL Combined X-ray Dosimeter (CXD) energetic charged particle sensors. Details of the CXD-equipped platforms, including naming conventions, data coverage and caveats can be found in Morley et al. (2017) and Morley et al. (2023). The data release includes energetic electron flux, among other data products, at a four-minute temporal resolution, with almost 300 satellite-years of CXD data starting from early 2001 (Morley et al., 2023). The public data is archived by National Oceanic and Atmospheric Administration (NOAA) National Centers for Environmental Information (NCEI) and indexed at the US government’s data.gov portal (Morley et al., 2017). The data files are provided as plain-text (ASCII), with a self-describing header containing both variable metadata and a file contents description, specified using a JavaScript Object Notation (JSON) header. The JSON-headed ASCII format has previously been used by the NASA Van Allen Probes mission magnetic ephemeris products, and is supported by community software including SpacePy (Morley, Welling, et al., 2010; Niehof et al., 2022) and Autoplot (Faden et al., 2010).

The CXD instrument includes three sub-systems that measure energetic particles (e.g., Carver et al., 2020). CXD is comprised of the Low-Energy Particle (LEP) detector, which measures electrons from  $\simeq 120$  keV to  $\gtrsim 1$  MeV in five energy channels (cf. Tuszewski et al., 2004, but note that the lower threshold can be commanded to either 120 or 140 keV), and two High-energy X-ray and Particle sensors, HXP1 and HXP2 (Carver et al., 2020) that provide an additional six energy channels covering energies upwards of  $\simeq 1.3$  MeV (e.g., Morley et al., 2014, 2016). Measurements are taken with a commandable accumulation interval that is set to 240 seconds for normal operation. Omnidirectional fluxes are calculated using a forward model described in Morley et al. (2016), employing a flexible sum of several basis functions designed to capture the sometimes-complex energy spectra observed in the inner magnetosphere. We note that due to the limited number of energy channels across a wide range of energies, the fits are necessarily smooth and cannot capture very narrow features.

The fitted omnidirectional electron fluxes have been validated through on-orbit cross-calibration with Van Allen Probes (Morley et al., 2016) and subsequent analyses (e.g.,

Olifer et al., 2024). While the omnidirectional electron flux has been shown to be generally consistent with Van Allen Probes measurements at energies up to  $\sim 4$  MeV (Morley et al., 2016), these data products have primarily been used in the study of radiation belt dynamics (Boynnton et al., 2017; Olifer et al., 2019; George et al., 2022; Rodger et al., 2022; Kalliokoski et al., 2023). In this work we focus more heavily on the lower energies measured by CXD to demonstrate the value of LANL-GPS data in monitoring substorm injections inside geosynchronous orbit.

One unique feature of the LANL-GPS data set is the large constellation of instruments, as the GPS satellites are distributed across six orbital planes, each near-circular and inclined at a nominal 55 degrees. The satellites typically cross the magnetic equator near  $L=4$  and map to higher  $L$  through their orbit, allowing sampling from  $L\sim 4$  to open field regions (cf. Morley et al., 2016, Figure 3). Thus, by combining data from multiple platforms it is possible to monitor a wide range in  $L$  simultaneously (e.g., Morley, Friedel, et al., 2010; Olifer et al., 2019; George et al., 2022). However, it is also important to note that the six orbital planes are roughly evenly spaced in local time (e.g., Morley et al., 2016), thus providing the ability to examine the evolution of electron fluxes in different local time sectors, on timescales longer than the 240-s resolution of the CXD data product.

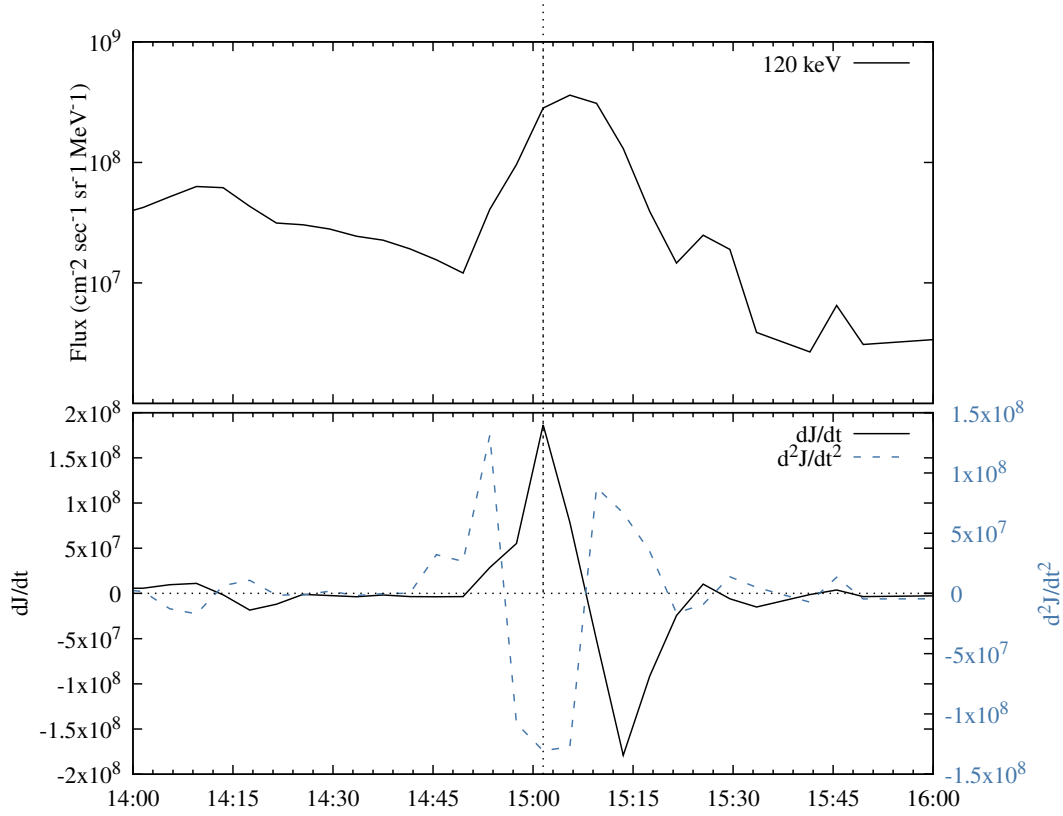
### 3 Methodology

The events identified in this work were initially identified using ancillary data sources, so approximate injection times were already known. Here we provide a quantitative algorithm for identifying the injection time from the LANL-GPS energetic particle data, and we note that extending this to “unsupervised” injection identification (e.g., for cataloguing injections across the full data set) is ongoing but beyond the scope of this work.

First, data for the entire GPS constellation active during the two days surrounding the injection time is collected (see Section 7). GPS particle data in the form of JSON-headed ASCII files is loaded using routines included in the SpacePy library (Morley, Welling, et al., 2010; Niehof et al., 2022). These data are then down-selected to include only those platforms which were at  $L \leq 6$  at the time of the injection. The remaining platforms are then organized by local time position at the time of the injection.

The canonical view of injections as described by Mauk and Meng (1983) and others has the injection originating near midnight and being deflected either East (dawnward) or West (duskward) depending on charge state by the combination of gradient and curvature drift in the approximately dipolar magnetic field. Thus we expect electrons injected to be observed first near midnight and then later at increasing distance eastward, i.e., at increasing local time. Time series plots of electron energies  $< 1$  MeV are generated and combined in a stack plot with the order determined by increasing local time of the associated GPS platform at the time of the injection. This visualization technique allows for rapid identification of the GPS platform closest to injection onset. Visual inspection also permits rapid bounding of the electron injection boundary based on local time locations of platforms showing dispersionless injection signatures (i.e., all observed energies increase simultaneously) versus those showing a dispersed injection (i.e., higher energy particle enhancements appear before those at lower energies).

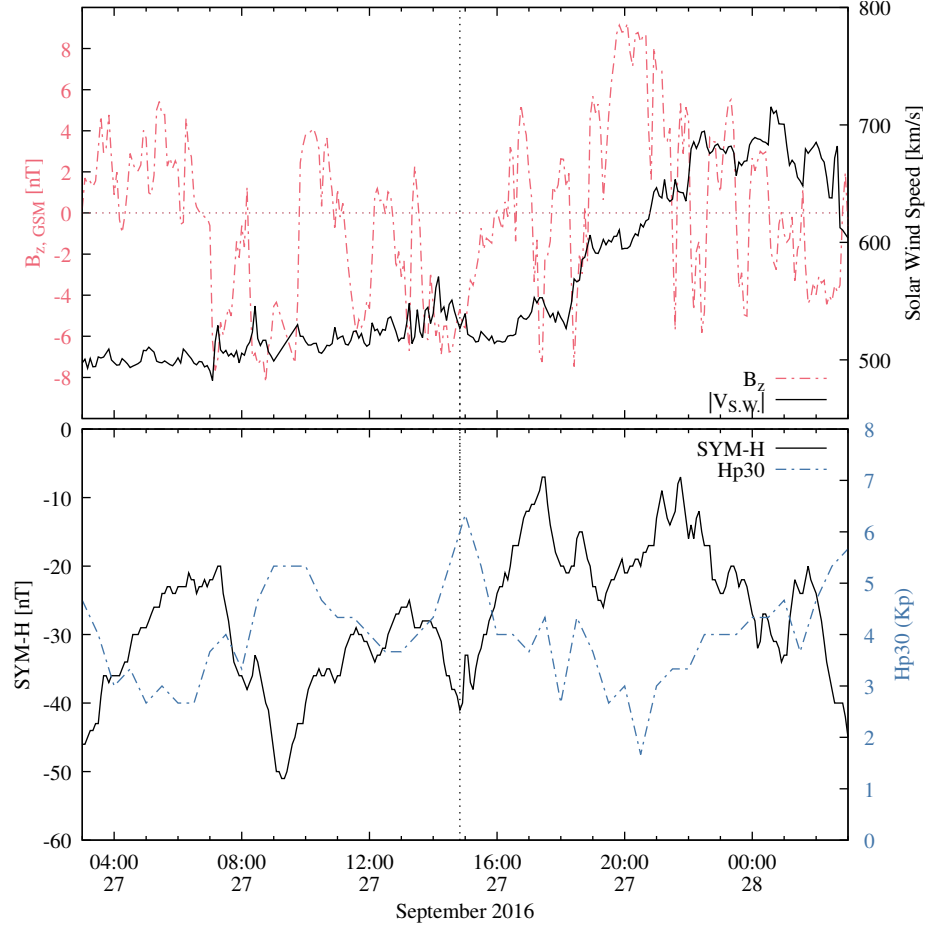
Injection timing at a given platform and energy was determined by finding the mathematical inflection points of time series flux values near the injection time. Discrete  $\frac{dJ}{dt}$  and  $\frac{d^2J}{dt^2}$  values were calculated for fluxes  $J$  at reported energies  $< 1$  MeV at platforms with dispersed injection signatures. Local maxima in  $\frac{d^2J}{dt^2}$  preceding local maxima in  $\frac{dJ}{dt}$  were identified as the ‘onset’ inflection point and form the basis for all injection timing in this work. This technique is demonstrated in Figure 1.



**Figure 1.** (top) Omnidirectional flux as a function of time during an electron injection at some energy. (bottom) The first and second derivative of the omnidirectional flux. See text for details.

As the injection boundary is, by definition, the point at which particles are no longer being actively energized by the injection mechanism and begin to drift adiabatically, the last location where particles associated with a dispersed injection signature, but with different energies, appear simultaneously should be at the injection boundary. To determine this location, we employ the SHIELDS particle tracing model (PTM; Morley et al., 2021; Brito et al., 2018; Woodroffe et al., 2018; Jordanova et al., 2018) to track sample electrons observed outside the injection boundary through a magnetic field model to find their common nexus. For this study we have employed the TS04 magnetic field model (Tsyganenko & Sitnov, 2005) implemented using the LANLGeoMag software suite (Henderson et al., 2024). Single test particles were traced backwards in time from the location and time the injection was observed as part of a dispersed injection at a given platform. Particles were assumed to be locally mirroring (i.e., a local pitch angle of  $90^\circ$ ) at the time of observation. While the electric field is important for particle drifts at lower energies, it is commonly omitted for study of electrons with energies  $\geq 100$  keV (e.g., Huang et al., 2023). While SHIELDS-PTM can include an electric field in solving for the particle motion, including a global electric field that is self-consistent with the magnetic field model is non-trivial for the case of TS04 and here we choose to neglect this effect.

Due to uncertainties in both energy and timing determination in GPS particle data, arising largely from the flux forward modeling and the temporal resolution (see section 2), intersections between different test particle energies launched from the same platform occur at a broad range of time and local time location. This makes boundary location



**Figure 2.** A summary of solar wind conditions (top) and geomagnetic indices (bottom) during the 24 hours surrounding the 2016-09-27 event.

identification by visual inspection of drift traces, the method utilized by Reeves et al. (1991), challenging. To mitigate this challenge we report the mean and associated standard deviation of all intersection locations in Coordinated Universal Time (UTC) and local time when estimating the eastward injection boundary location (see Figures 5 and 9 below).

## 4 Event 1: 2016-09-27

### 4.1 Global Overview

On September 27th of 2016 a fast solar wind (black line, top, Figure 2) coupled with relatively large magnitude fluctuations about zero in the z-component of the interplanetary magnetic field (red line, top, Figure 2), both obtained from the OMNI dataset (King, 2005), led to notable geomagnetic activity. Ring current indices such as SYM-H (black line, bottom, Figure 2) reached a minimum of  $\approx -52$  nT, indicating a moderate geomagnetic storm (Loewe & Prölss, 1997). Simultaneously the Hp30 index (Yamazaki et al., 2022)(essentially a 30-minute resolution Kp index, blue dashed line, bottom, Figure 2) shows at least two discrete intervals of enhanced activity, indicated by values of Hp30  $> 6$ . Each of these enhancements is coincident with a positive excursion in the SYM-

H index, indicating a likely connection to substorm injection (e.g., Pulkkinen et al., 2003). For our first event we focus on the second of these enhancements, seen in Figure 2 at approximately 1500 UTC, which was previously reported as a substorm injection observed at Van Allen Probes by H.-J. Kim et al. (2021).

## 4.2 Data

The energetic charged particle environment for this event was also observed by the GPS constellation. Figure 3 shows the approximate locations of the GPS platforms at the time of the substorm, approximately 1500 UTC, referenced from Figure 2. Note the reasonably uniform coverage both in local time and L-shell provided by the GPS constellation. This is in marked contrast to observations made by H.-J. Kim et al. (2021) using NASA’s Radiation Belt Storm Probes (RBSP) at only two locations (magenta dots, Figure 3) or LANL-GEO platforms which provide good coverage in local time, but are limited to geosynchronous orbit (*i.e.*  $L \approx 6.6$ , not shown).

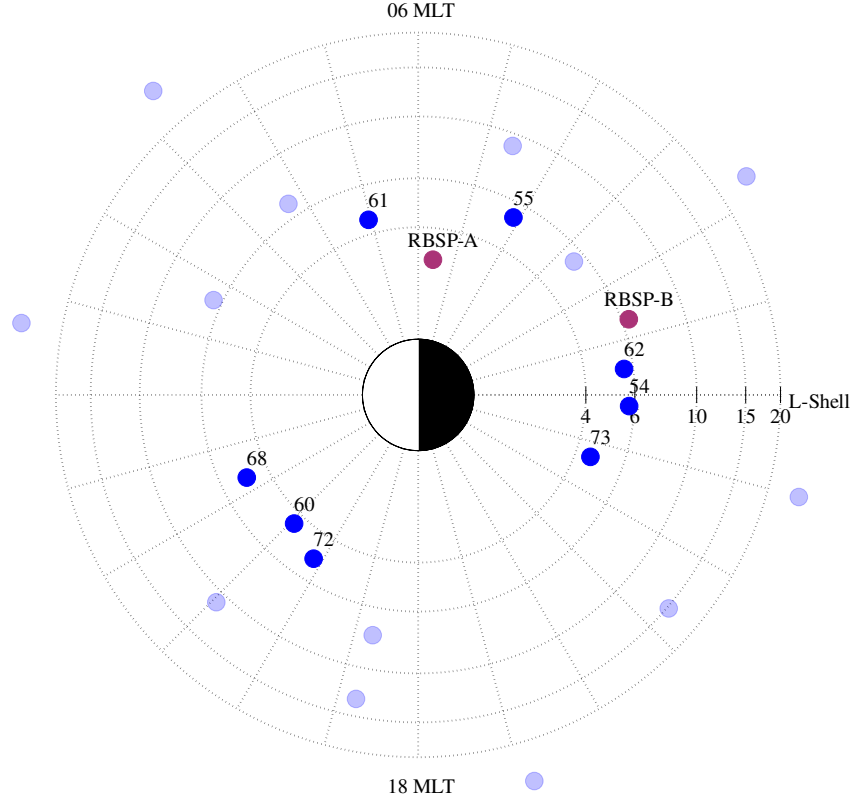
Nine GPS platforms were at low L (*i.e.*  $L < 6R_E$ ) at the time of the substorm. However ns57, located at a local time of approximately 2:45, rapidly moved out of the radiation belts afterwards and so is not included in this analysis. The remaining eight GPS platforms (ns54, ns62, ns55, ns61, ns68, ns60, ns72, and ns73) provide excellent azimuthal coverage of the outer radiation belt inside of GEO. Figure 4 shows the time series omnidirectional flux of electrons at energies between 120 and 800 keV observed at the eight GPS platforms at low L. Graphs in Figure 4 are organized by location in local time at the beginning at the injection. Vertical dashed lines are added on each graph to indicate the earliest inflection point, typically at the highest energy analyzed, at the rise in flux denoting the electron injection at that platform. A solid black vertical line shows the onset inflection time for the first platform to observe the injection; in this case ns54.

At the start of the injection ns54 is located at  $L \approx 5.25 R_E$  and approximately 23:50 local time. ns62, at slightly lower L and just post-midnight, observes the initial injection essentially simultaneously with ns54 (*i.e.*, within the overlapping  $\pm 120$  s measurement window). The peaks of these injection signatures are also dispersionless within the intrinsic timing uncertainties of the data. The injection is then observed at ns55, located at around 03:45 local time and with the leading inflection point delayed more than 120 s from that observed by ns54. ns55 also observes an essentially dispersionless injection. In contrast, ns61 is located at approximately 07:00 local time and shows a strong time-energy dispersion relationship starting within the measurement window of ns55.

The degree of time-energy dispersion increases moving eastward (down in Figure 4), as expected by electrons drifting away from an injection acceleration region. The greatest time-energy dispersion is observed by ns73 located at approximately 23:00 local time. As such the eight GPS platforms with data shown in Figure 4 illustrate the circumnavigation of a single electron injection around the Earth inside of geosynchronous orbit.

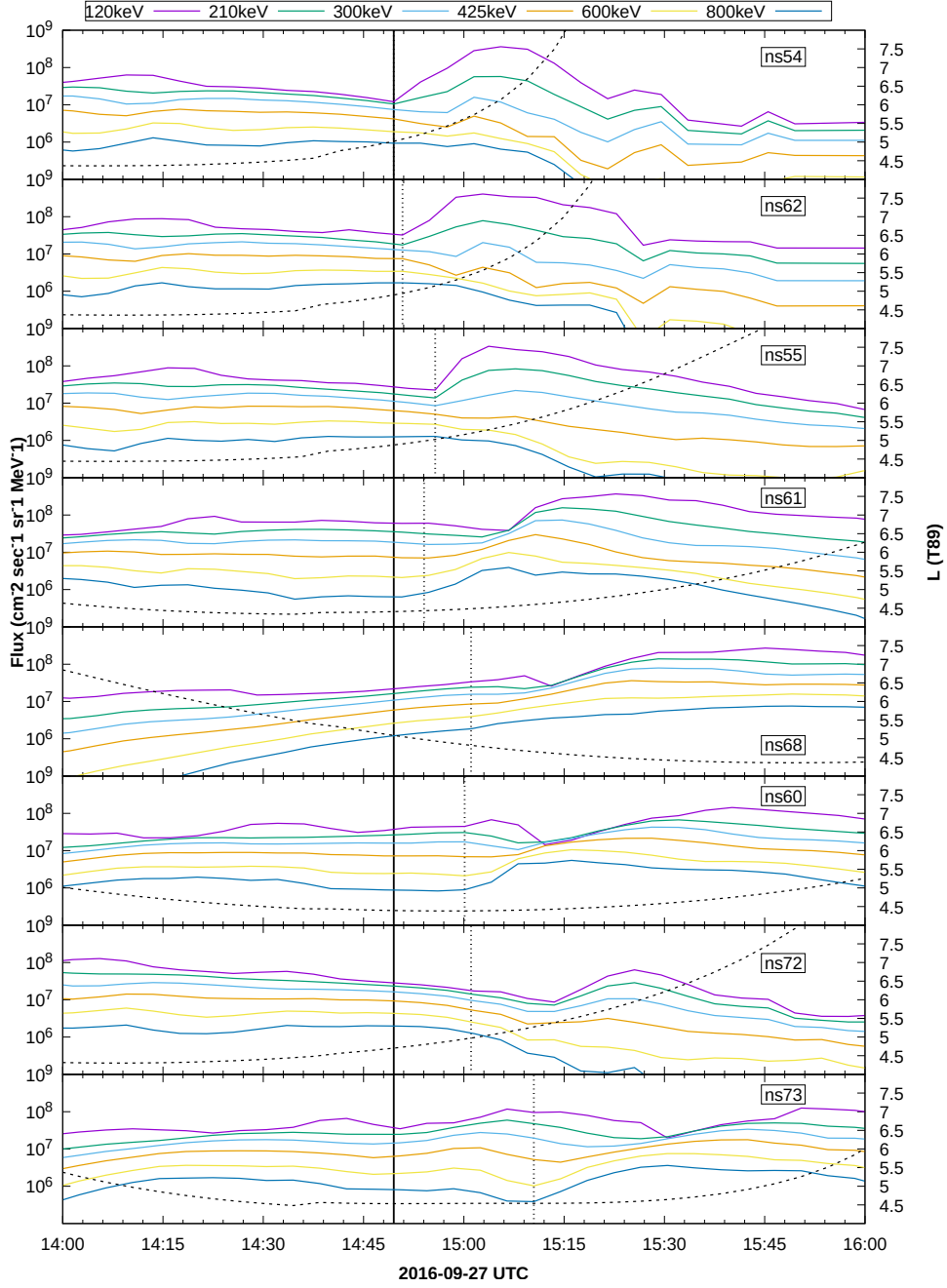
The broad azimuthal distribution allows us to make some inferences regarding the electron injection boundary (Birn et al., 1998). The western (duskward) boundary of the injection region can be placed with some confidence between ns54 and ns73 (*i.e.*, between 23:00 – 23:45 local time) by observing the timing and time-energy dispersion signatures of each. That is, the dispersionless nature of the injection observed by ns54 lends confidence to the interpretation of it being within (or just Eastward of) the injection region. On the other hand, the later onset and highly dispersed injection signature observed by ns73 indicates that it was outside the injection boundary and only observed drifted electrons after they had exited the injection boundary on the eastern side.

The eastern (dawnward) boundary of the injection region may be estimated using the same approach. Simultaneous observations of both an apparent dispersionless (ns55)

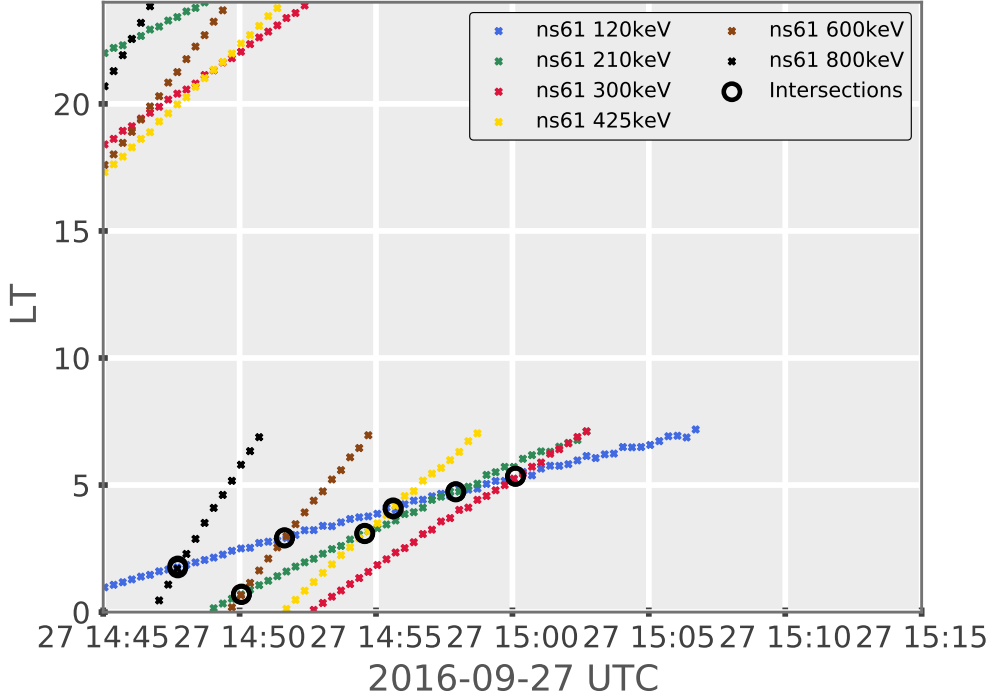


**Figure 3.** The locations of GPS platforms at approximately the beginning of the injection under study (2016-09-27T05:00:00) given in local time and L-shell coordinates. The dark blue markers indicate GPS platforms that were at  $L \leq 6$  at the time of injection, with the satellite vehicle number given as a text annotation. The light blue markers indicate the other CXD-equipped GPS platforms at this time. The locations of the Van Allen Probes are indicated by the magenta markers. Note that L-shell coordinates are given on a logarithmic radial scale. The Sun is to the left, as indicated by the shaded circle.





**Figure 4.** Time series omnidirectional electron flux at energies denoted by color (as indicated in the legend at the top of the figure). Platforms near midnight are placed at the top and then are located lower with increasing local time. Dotted lines are scaled to the right-hand vertical axis and show approximate L-shell at that time. A solid vertical line denotes the earliest signature of the injection observed at ns54; vertical dotted lines denote the earliest injection signature at that platform.



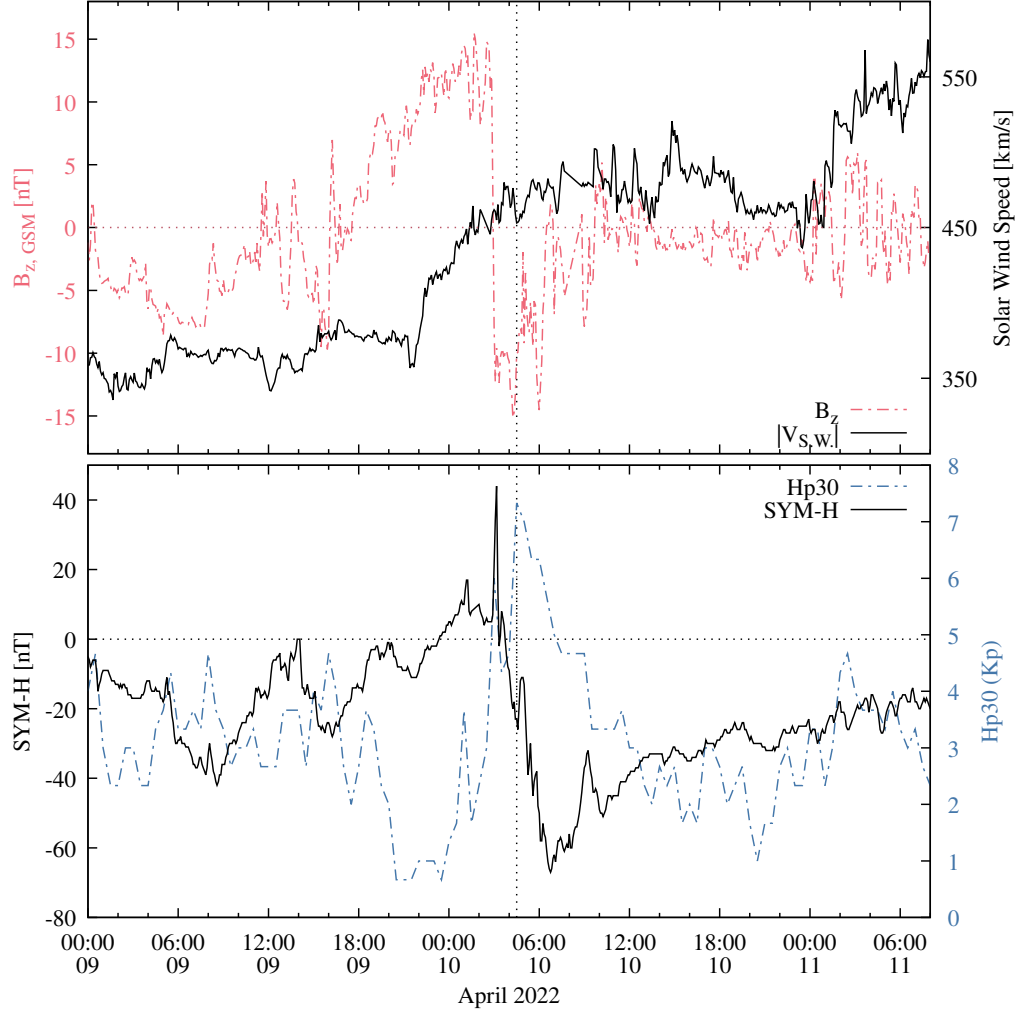
**Figure 5.** Drift paths of test electrons launched from ns61 at various energies listed in Figure 4. Black circles denote intersections in drift paths.

and dispersed (ns61) injection at adjacent GPS platforms indicates that the injection boundary is likely to lie between them. Note that due to the temporal resolution, it is possible for a platform just to the East of the injection boundary to observe an apparently dispersionless injection, thus the eastern extent of the injection boundary could also lie slightly to the West of ns55. As shown in Figure 3, this condition still leaves an approximately three hour window in local time in which the boundary could be. Any attempt to further refine the location of the eastern injection boundary will require a more subtle technique.

### 4.3 Particle Tracing Boundary Estimation

The backwards tracing method described in Section 3 is the technique we have chosen. Figure 5 shows the location in local time as a function of UTC of a selection of test electrons of various energies and launched backwards within the simulation from ns61. As discussed above, the initial launch location and time used is taken from the initial (i.e., rising) inflection point of omnidirectional flux at that energy and at that platform.

As seen in Figure 5, visual inspection supports an eastward boundary between 02:00 and 05:00 local time and, very approximately, 14:55 UTC, near the time and location of the injection as observed by ns55. We additionally note that the injection times identified for 210 keV and 300 keV have the same nominal time. This is due to the temporal resolution of the LANL-GPS data product described previously and an uncertainty of  $\pm 120$  s is implied for the initial position of each test particle trajectory. A total of 7 intersections occurred during the simulation time and on the dawn side (i.e., between 00 and 12 hours local time). The mean location in local time was  $3.24 \pm 1.52$  hours at  $14:53:57 \pm 4:04$



**Figure 6.** An overview of the solar wind and geomagnetic index parameters surrounding the 2022-04-10 event. Formatted after Figure 2.

UTC. This location is close to what was determined using other techniques discussed previously and overlaps the quasi-dispersionless observations at ns55.

It should be noted that additional test particles to represent observations from additional platforms were simulated in an effort to improve the overall statistics of the injection boundary estimate above. However, the additional uncertainties from the use of a semi-empirical magnetic field model and the omission of electric field effects are much larger with longer drift trajectories. We found that the additional uncertainties were cumulative, rather than random, and have therefore limited our analysis to test particles launched from the first platform with a clear dispersed injection signature.

## 5 Event 2: 2022-04-10

### 5.1 Global Overview

The GPS constellation observing an energetic electron injection was not unique to the September 2016 event discussed above. Another interesting example was observed at 2022-04-10T04:30 UTC. An overview of the solar wind conditions as observed in OMNI data (King, 2005) surrounding this event are shown in Figure 6. The center of a large flux rope containing highly compressed plasma impacted the Earth at approximately 03:50 UTC and was quickly followed by a moderate geomagnetic storm and enhanced Hp30 ( $> 7$ ). The peak in Hp30 again aligns with a positive excursion in the SYM-H index, indicating a likely substorm injection. This interval was previously studied by Nagai et al. (2023) who noted the occurrence of a large dipolarization at about 04:30 UT, as well as a mid-latitude positive bay measured on the ground. The combination of signatures, in the presence of an injection observed by LANL-GPS, strongly suggests that this is a substorm injection (e.g., Haiducek et al., 2020).

### 5.2 Data

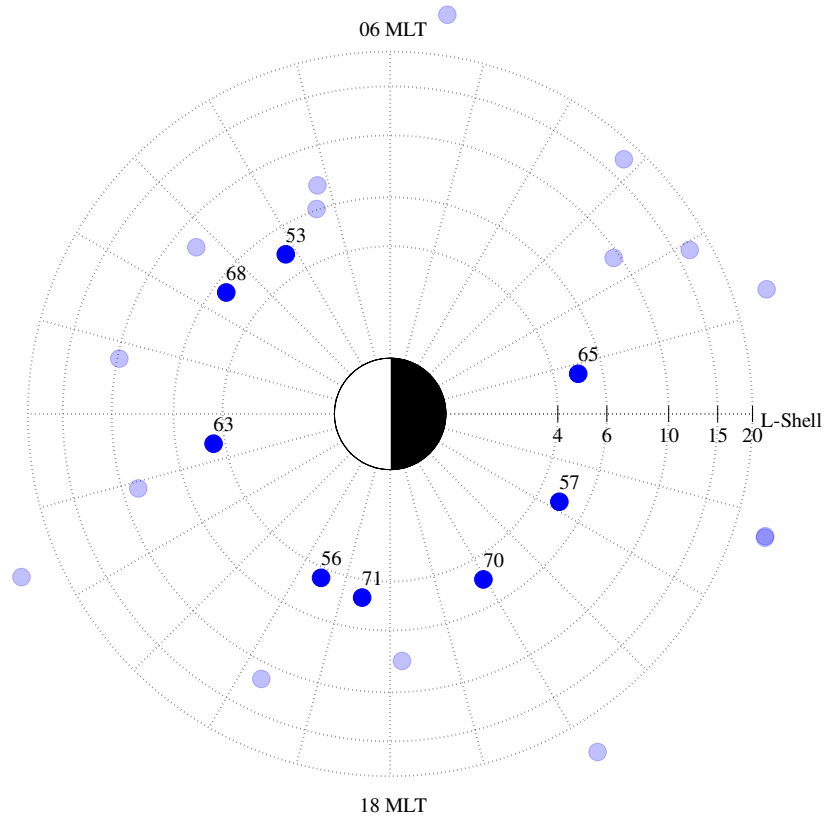
Figure 7 shows the position of the GPS constellation near the initial observation of the energetic electron injection. It is worth noting that the Van Allen Probes are not shown in Figure 7 due to their having been decommissioned and deorbited in 2019. Figure 8 shows the omnidirectional electron flux as a time series for those GPS platforms at low L in increasing local time from top to bottom.

The injection appears dispersionless at ns65 near 00:30 local time and ns57 near 22:00 local time, with the initial rising inflections at each of these platforms within 120 s of each other (i.e., simultaneous within measurement uncertainty). On the westward (duskward) side, differences in initial time-energy dispersion signature between ns57 and ns70 allow us to confidently place the injection boundary on that side between these two platforms in local time at approximately 21:00. This technique is less useful in locating the eastern boundary given the approximately eight hour difference in local time between ns65 at 0.89 LT and the next platform at low L to the east, ns53 at 8.40 LT, where the injection signature is highly dispersed. Thus we need to use another member of the GPS constellation to help constrain the extent of the injection region.

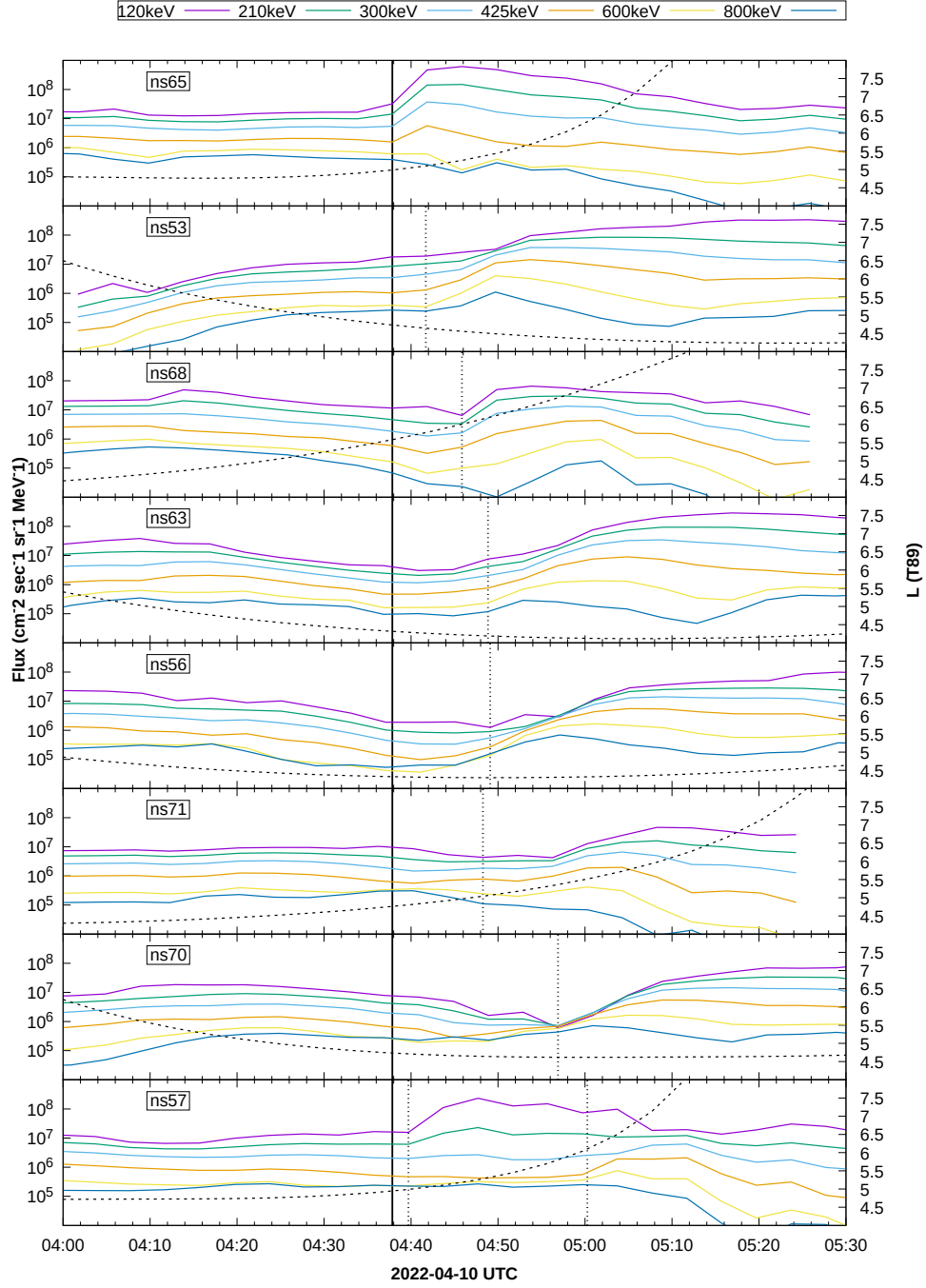
A closer look at the omnidirectional flux time series data for ns57 during this event shows an interesting feature. After the initial dispersionless injection in lower energy channels at approximately 04:40 UTC, a second set of injection-like flux increases are observed at higher energies at approximately 5:00 UTC. This second set of injection signatures display a large time-energy dispersion and are observed after those seen at ns70 further westward. The timing and dispersion relation of this second 'injection' are appropriate if they represent the drifted high-energy electrons, or drift echo, of the original injection after circumnavigating from the eastern injection boundary (Mauk & Meng, 1983). Thus a single platform observes not only the initial injection from within the injection acceleration region, but also the drifted particles after the dispersionless injection itself has passed.

### 5.3 Particle Tracing Boundary Estimation

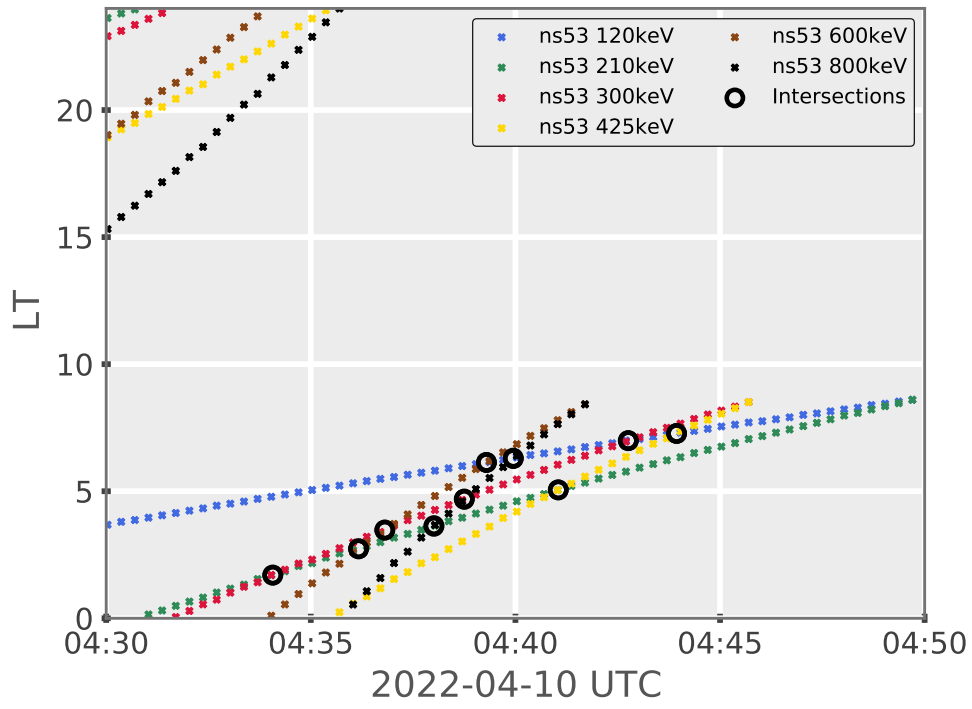
To constrain the likely extent of the injection region, which could not be determined directly from the LANL-GPS data, we again apply the backwards-time test particle tracing technique discussed previously. The results, utilizing data from ns53, are shown in Figure 9. In this case there are 10 drift path intersections which can be analyzed. The mean local time location of the eastern boundary is estimated to be  $4.80 \pm 1.79$  hours at 04:39:36 $\pm$ 2:52 UTC. All trajectory intersections lie between ns65 and ns53, as expected, and we note that while the latest candidate boundary locations implied by the test par-



**Figure 7.** Locations after Figure 3 of GPS platforms at the initial observations of the electron injection observed by GPS on 2022-04-10.



**Figure 8.** Time series data of omnidirectional electron flux for those GPS platforms at  $L < 6$  at the initial injection observation. Format follows Figure 4.



**Figure 9.** Drift paths of test electrons launched from ns53 at various energies listed in Figure 8. Formatted after Figure 5.

332 ticle trajectories are closer to 7 LT the  $\simeq 4$  minute uncertainty arising from the timing  
 333 resolution of the LANL-GPS data corresponds to approximately 1 hour of local time for  
 334 the 120 keV electrons and  $\geq 3$  hours of local time for 600 keV electrons. These uncer-  
 335 tainties are consistent with the reported standard deviation of intersection locations. Even  
 336 with this uncertainty, the test particle tracing has significantly constrained the injection  
 337 extent.

## 338 6 Discussion

339 While substorm injections have previously been observed inside geosynchronous or-  
 340 bit (Lopez et al., 1990; Turner et al., 2015; H.-J. Kim et al., 2021), the key observations  
 341 have previously relied on satellites in highly-elliptical orbits such as AMPTE-CCE (e.g.,  
 342 Lopez et al., 1990) or Van Allen Probes (Turner et al., 2015). These missions have had  
 343 limited lifespans, typical of scientific missions in the inner magnetosphere. The substorm  
 344 injections presented above illustrate that operational spacecraft, forming a constellation  
 345 in Medium-Earth Orbit, can identify and characterize substorm injections.

346 Using the LANL-GPS electron data we have shown that typical dispersionless and  
 347 dispersed injection signatures are seen at low L when the GPS satellites are near the mag-  
 348 netic equator. The large constellation allows us to directly constrain the injection region  
 349 from observations alone as well as where the data only weakly bound the injection re-  
 350 gion we have shown that we can augment the observations with test particle simulations.  
 351 The Eastward extent of the electron injection region was identified for two case studies:  
 352 2016-09-27 (Event 1), and 2022-04-10 (Event 2). In event 1 the Eastward extent was bounded  
 353 by a nominally dispersionless injection signature at 3.91 LT and a clearly dispersed sig-  
 354 nature at 6.93 LT. The boundary estimated using backwards-time particle tracing was  
 355  $3.24 \pm 1.52$  LT. In event 2, the Eastward extent was bounded by a nominally disper-  
 356 sionless injection signature at 0.89 LT and a clearly dispersed signature at 8.40 LT. The  
 357 boundary estimated using backwards-time particle tracing was  $4.80 \pm 1.79$  LT. Thus,  
 358 in addition to observing electron injection at low L we can use even a limited portion  
 359 of the GPS constellation to bound the injection region.

360 The electron injection boundaries estimated here are significantly offset towards  
 361 dawn compared to those suggested by the originators of the injection boundary model  
 362 (e.g., Mauk & McIlwain, 1974; Konradi et al., 1975; Mauk & Meng, 1983). Somewhat  
 363 more recent work by Birn et al. (1998) supports an offset electron injection boundary  
 364 to the East of midnight, while ion injections would be offset to the West of midnight.  
 365 Still, the eastward boundaries estimated here are at significantly greater local time than  
 366 had been previously discussed in literature. Two possibilities occur to us: first, previ-  
 367 ous investigations of this offset were made at geosynchronous orbit (i.e.,  $L \approx 6.6$ ), fur-  
 368 ther from Earth than the current study. This would imply a greater amount of time for  
 369 magnetic field gradient and curvature forces to influence the bulk plasma as it moves Earth-  
 370 ward, possibly pushing electrons closer to dawn than is nominally observed at GEO. This  
 371 possibility is supported by the LANL-GEO observations reported by H.-J. Kim et al. (2021),  
 372 which show a dispersionless electron injection at a local time of  $\approx 19.4$ , much further  
 373 westward than was observed by GPS during the same event. A second possibility is that  
 374 the measurement uncertainties inherent in the GPS particle data do not permit sufficient  
 375 resolution of time-energy dispersion to estimate the boundary accurately. Moving only  
 376 a single standard deviation toward midnight from the mean eastern boundary value from  
 377 the two events discussed above would produce eastern boundaries at 3.63 and 4.67 lo-  
 378 cal time respectively, in much better agreement with studies performed at GEO.

379 Differences in the injection time reported by H.-J. Kim et al. (2021) point to an-  
 380 other possible source of uncertainty. Kim et al. report the injection on September 27,  
 381 2016 occurring at  $\sim 14:57$  UTC, 7-8 minutes after the time we report here. This is po-  
 382 tentially explained by our use of the rising ‘onset’ inflection while the precise timing method



used by Kim et al. is not made clear. Existing unsettled conditions in the outer radiation belt may influence the precising timing of the calculated rising inflection point, adding to the overall uncertainty. Other methods, such as using the maximum  $\frac{dJ}{dt}$  for a given energy or the ‘cresting’ inflection (local minima in  $\frac{d^2J}{dt^2}$ ) may prove to be more precise, although comparison between methods over more than two events would be necessary to determine this. Regardless, we feel that the repeatability of a mathematically precise definition such as inflection point or maximum slope is preferable to visual inspection for timing analysis.

It is worth noting that both of the events discussed here occurred during substorms with Hp30 index values greater than 6. This is consistent with the Kp-dependence of the radial penetration of the injection boundary (e.g., Mauk & McIlwain, 1974; Lopez et al., 1990), which would imply that an enhanced Kp (or Hp30) is a prerequisite for observing a substorm particle injection at low-L. However, while consistent, the small sample does not allow us to draw conclusions beyond the consistency. However, the LANL-GPS data provides a clear opportunity to examine the conditions required for substorm injections to penetrate to low L as it provides a long baseline, multi-platform dataset that has now been demonstrated to be suitable for identifying substorm injections. Development and validation of an automated procedure for identifying injections would allow the cataloguing of a statistically significant sample of injections that could be used to understand the necessary and sufficient conditions for injections to low L.

The GPS constellation has significantly longer operating lifespan than inner magnetosphere science missions, making the capability to observe substorm injections inside geosynchronous orbit extremely valuable for scientific study. As the GPS platforms are already a constellation measurement, we note that combination of GPS with other operational missions (LANL-GEO, GOES) represents an ongoing opportunity for multi-point study previously only possible using serendipitous observations from science missions (Turner et al., 2017). Serendipitous conjunctions with previous missions such as Van Allen Probes can be re-examined in the context of this operational super-constellation, and upcoming missions in geosynchronous transfer-like orbits, such as the Experiment for Space Radiation Analysis (ESRA; Maldonado et al., 2023) or GTOsat (Blum et al., 2020) will be afforded global context for substorm injection or broader radiation environment dynamics.

## 7 Conclusion

We present the first observations of substorm energetic electron injections from LANL-GPS instrumentation at Medium-Earth Orbit, thus demonstrating a monitoring capability that has previously been overlooked. We show two examples of substorm injections observed at low L ( $\simeq 4.5$ ), and using a new, easy-to-implement, quantitative method for identifying the onset time of the injection we illustrate how distributed measurements from an operational constellation can be used to bound the injection region.

Although the CXD instrument does not provide measurements as exquisite as, for example, the ECT suite on Van Allen Probes (Spence et al., 2013), this work shows that operational LANL-GPS data is a valuable resource for injection studies. The constellation provides sufficient spatial coverage to reliably observe injections that penetrate inside geosynchronous orbit. The temporal resolution is still sufficient to differentiate between approximately dispersionless and dispersed injection signatures within a couple of hours of local time. Where spatial coverage alone does not adequately constrain the substorm injection region, we show that backwards-time test particle tracing can help bound the injection.

The operational nature of the GPS constellation also brings the benefit of a long mission lifetime: CXD instrumentation, for example, has been continuously flown since

2001. In combination with the relatively large number of platforms able to simultaneously observe any given event, for most events LANL-GPS should be placed to observe both dispersionless and dispersed injection signatures. When higher-fidelity scientific measurements are available, the GPS constellation provides key context regarding the global-scale injection event. However, in the absence of flagship radiation belt missions, GPS provides a capability for monitoring and characterizing substorm injections at low L. Upcoming missions such as ESRA (Maldonado et al., 2023) or GTOsat (Blum et al., 2020) will provide opportunities for combining the LANL-GPS constellation data with higher-fidelity point measurements for studying inner magnetospheric radiation dynamics.

## Open Research Section

GPS data (v1.10) is available from NOAA (<https://www.ncei.noaa.gov/products/GPS-energetic-particles>). The OMNI data, including solar wind parameter and geomagnetic indices, were obtained from the GSFC/SPDF OMNIWeb interface (<https://omniweb.gsfc.nasa.gov>). Analyses and plotting was primarily performed using the open-source SpacePy (Niehof et al., 2022; Morley et al., 2024) software library, the LANLGeoMag library (Henderson et al., 2024), the SHIELDS-PTM code (Morley et al., 2021), and GNUplot (<https://gnuplot.info>).

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